

Data Cleaning & Preparation

- Ensured Odometer and price columns were present and loaded.
- Outlier Removal: filter out the outliers that could confuse the ML learning models
- Used `.dropna()` to ensure that the linear and ridge regression won't break

Feature Engineering

- **Temporal Logic (Age):** Instead of using raw year, we created an age feature ($\$2026 - \text{year}$). Models understand "5 years old" much better than "2021."
- **Interaction Features:** We created miles_per_year to give the model context. A low-mileage car for its age is a much stronger price signal than just total mileage alone.
- **Categorical Encoding:** Since models can't "read" words, we used **One-Hot Encoding** (`pd.get_dummies`) to turn brands like Toyota or Ford into binary numbers (0s and 1s).

Visualization Process

- Used Histograms to see if prices were skewed. This helped decide if we needed to filter out the extreme.
- We used line plots for depreciation. The visual proved that car values don't drop in a straight line, it tends to be random, which then made sense to use polynomial regression and random forest
- Brand Comparison: we used boxplots to see the price floor and ceiling for different manufactures. This confirmed that the brand of car plays an important part for predicting the value.
- Correlation Mapping: we used heatmap to see which features have a strong relation towards the price. This allowed us to get rid of useless columns and focus on high impact ones like the age and year of the car

Test different k values in order to use knn